

Homework 8

Problem 1

Consider the general problem of estimating a population proportion p for some binary (success/failure) categorical variable of interest in a large population. Select a sample of size n and let X be the number of successes in the sample. (We are considering a general problem here, but recall your particular context from a previous homework.)

We'll consider two estimators of p :

$$\begin{array}{ll}\text{Sample proportion:} & \hat{p} = \frac{X}{n} \\ \text{Plus four:} & \tilde{p} = \frac{X + 2}{n + 4}\end{array}$$

Suppose we observe $x = 6$ successes in a sample of size $n = 10$. For this sample $\hat{p} = 0.6$ and $\tilde{p} = 0.571$.

1. Compute and interpret the probability of observing $x = 6$ successes in a sample of size $n = 10$ if $p = 0.6$.
2. Compute and interpret the probability of observing $x = 6$ successes in a sample of size $n = 10$ if $p = 0.571$.
3. Which of the estimates, 0.6 or 0.571, is a *better estimate* of p based on a sample of size $n = 10$ with $x = 6$ successes? That is, which estimate, 0.6 or 0.571, is closer to the true p ? Explain.
4. Which of the estimates, 0.6 or 0.571, would you choose as your point estimate of p based on a sample of size $n = 10$ with $x = 6$ successes? Explain. Hint: consider parts 1 and 2.
5. Suppose you observe $x = 6$ successes in a sample of size $n = 10$. Carefully write the likelihood function. Hint: what is the distribution of X ?
6. Use any software of your choice to plot the likelihood function. Based on your plot, what value appears to be the maximum likelihood estimate of p based on this sample? (You can do this just by zooming in on the plot.)
7. Carefully write the log-likelihood function.
8. Use calculus to find the maximum likelihood estimate of p based on $x = 6$ successes in a sample of size $n = 10$.

Problem 2

Continuing Problem 1, now consider a general sample of size n with x successes.

1. Carefully write the likelihood function. (The expression should contain x and n .)
2. Continuing the previous part, carefully write the log-likelihood function. (The expression should contain x and n .)
3. Use calculus to find a general expression for the maximum likelihood estimate of p . Your answer should be an expression involving x and n .

Problem 3

Assume a $\text{Poisson}(\mu)$ model for the number of home runs hit (in total by both teams) in a MLB game. Let $X_1, \dots, X_n$ be a random sample of home run counts for n games. (The context doesn't matter, but it helps to have something in mind rather than just the general problem.)

Suppose we want to estimate $\theta = \mu e^{-\mu}$, the probability that any single game has exactly 1 HR (for $\text{Poisson}(\mu)$, $P(X = 1) = e^{-\mu} \mu^1 / 1! = \mu e^{-\mu}$). Consider two estimators of θ :

- $\hat{\theta} = \bar{X} e^{-\bar{X}}$
- $\hat{p}$ = sample proportion of 1s = $\frac{\text{number of games in the sample with 1 HR}}{\text{sample size}}$

1. Compute the value of $\hat{\theta}$ based on the sample (3, 0, 1, 4, 0). Write a clearly worded sentence reporting in context this estimate of θ .
2. Compute the value of $\hat{p}$ based on the sample (3, 0, 1, 4, 0). Write a clearly worded sentence reporting in context your estimate of θ .
3. Which of these two estimators is the MLE of θ in this situation? Explain, without doing any calculations.
4. It can be shown that $\hat{p}$ is an unbiased estimator of θ . Explain in words what this means.
5. Is $\hat{\theta}$ an unbiased estimator of θ ? Explain. (You don't have to derive anything; just apply a general principle.)
6. Suppose $\mu = 3.4$ and $n = 5$. Explain in full detail how you would use simulation to approximate the bias of $\hat{\theta}$ in this case. In particular, specify the spinner you would use to simulate values.
7. Explain in full detail how you would use simulation to approximate the bias *function* of $\hat{\theta}$ when $n = 5$.

Problem 4

Continuing Problem 3.

1. It can be shown that $\text{Var}(\hat{p}) = \frac{\theta(1-\theta)}{n}$. Compute $\text{Var}(\hat{p})$ when $\mu = 3.4$ and $n = 5$. Then write a clearly worded sentence interpreting this value.
2. Suppose $\mu = 3.4$ and $n = 5$. Explain in full detail how you would use simulation to approximate the variance of $\hat{\theta}$.
3. Explain in full detail how you would use simulation to approximate the variance function of $\hat{\theta}$ (if $n = 5$).

Problem 5

Continuing Problems 3 and 4

1. Compute $\text{MSE}(\hat{p})$ when $\mu = 3.4$ and $n = 5$. (You can do the next part first if you want, but it helps to work with specific numbers first.)
2. Derive the MSE function of $\hat{p}$. (Hint: use facts from previous parts.)
3. Suppose $\mu = 3.4$ (and $n = 5$). Explain in full detail how you would use simulation to approximate the MSE of $\hat{\theta}$.
4. Suppose $\hat{\theta}$ has smaller MSE than $\hat{p}$ when $\mu = 3.4$ (and $n = 5$). Which estimator would you prefer? Answer, but then explain why this information alone is not really useful.
5. Explain in full detail how you would use simulation to approximate the MSE *function* of $\hat{\theta}$ (if $n = 5$).
6. Suppose you determine that $\hat{\theta}$ has smaller MSE for almost all values of μ regardless of n . Recall parts 1 and 2 of Problem 3. Which value of these two estimates (the values from parts 1 and 2 of Problem 3) is a better estimate of θ ? That is, which of the two estimates is closer to the true θ ? Explain.
7. Suppose you determine that $\hat{\theta}$ has smaller MSE for almost all values of μ regardless of n . Recall parts 1 and 2 of Problem 3. Which value of these two estimates (the values from parts 1 and 2 of Problem 3) would you choose as your point estimate of θ ? Explain.